

BISHAL ROY

SENIOR SOFTWARE ENGINEER

Python Backend | FastAPI | Django | AI Platforms | Real-Time Trading Systems

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PROFESSIONAL SUMMARY

Python backend and full-stack developer with 3+ years of experience building scalable APIs, AI-driven education platforms, automation systems, and real-time trading applications. Experienced in FastAPI, Django, PostgreSQL, WebSockets, Docker, authentication, performance optimization, and production software.

CORE SKILLS

Backend	Python, FastAPI, Django, Flask, REST APIs, WebSockets
Database	PostgreSQL, SQLAlchemy, SQL, Redis
Security	JWT, OAuth, secure login, role-based access
DevOps	Docker, Git, AWS, Linux, CI/CD
AI / Data	Pandas, NumPy, TensorFlow, PyTorch, Matplotlib

PROFESSIONAL EXPERIENCE

Senior Software Engineer

Jan 2024 - Present

INFORMATICAЕ TECHNOLOGY PVT. LTD.

Kolkata, India

- Built AI-driven teacher modules for assignment generation and academic workflow automation.
- Reduced manual preparation effort by 85% and automated answer checking and student analytics.
- Built multilingual, multi-difficulty assessment configuration with weighted scoring logic.
- Mentored junior developers and improved code quality through reviews and optimization.

Software Engineer

Aug 2023 - Dec 2024

INFORMATICAЕ TECHNOLOGY PVT. LTD.

Kolkata, India

- Built scalable REST APIs using FastAPI for Hello Teacher AI.
- Implemented secure authentication and authorization using JWT and OAuth.
- Optimized SQL queries, indexing, and caching, improving API response time by 50%.
- Worked with Docker and CI/CD pipelines for reliable deployment and testing.

Backend Developer Intern

Mar 2023 - Aug 2023

RUNTIME SOLUTIONS

Kolkata, India

- Built scalable web applications using Django and Flask.
- Designed optimized relational database schemas and implemented Redis caching.

SELECTED PRODUCT WORK

KUBERAV - ALGORITHMIC TRADING PLATFORM

kuberav.com

Personal web-based trading platform for live market data, broker orders, strategy automation, paper trading, options, risk controls, and real-time WebSocket updates from one dashboard.

- Live and paper trading, broker integration, positions, order book, and trade history
- Stop-loss, targets, trailing exits, charts, option chain, and webhook automation
- FastAPI, Python, SQLAlchemy, PostgreSQL, REST APIs, WebSockets

HELLO TEACHER AI - AI EDUCATION PLATFORM

hello-teacher.ai

Contributed scalable backend APIs, secure authentication, teacher assignment workflows, subject management, and model question generation for an AI-powered education platform.

- Python, FastAPI, PostgreSQL, JWT, OAuth, REST APIs, Docker

ACADEMICAЕ SCHOOLS - ACADEMIC PLATFORM

aps.academicae.com

Contributed dynamic assessment configuration, automated answer checking, student analytics, multilingual content, difficulty logic, and academic workflow automation.

- Python, FastAPI, PostgreSQL, SQL, REST APIs, Docker

MEASURABLE IMPACT

85%

manual effort reduced

80%

grading time reduced

50%

API response improved

3+

years experience

EDUCATION

B.Tech in Computer Science Engineering

Gargi Memorial Institute of Technology, Kolkata | GPA: 8.5 | 2019 - 2023

CERTIFICATION

Advanced Certification in Applied Data Science, Machine Learning and IoT

Indian Institute of Technology Guwahati | Jan 2025 - Mar 2026

ADDITIONAL

Frontend

HTML, CSS, JavaScript, React

Focus

Backend architecture, performance, automation, and product engineering

Open to backend, platform, and senior software engineering opportunities.

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